METRIC

FF-B-171/37

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

BEARING, BALL, ANNULAR, DOUBLE ROW, ANGULAR CONTACT, FILLING SLOT, VERTEX OF CONTACT ANGLES OUTSIDE BEARING, ANSI/AFBMA DIMENSION SERIES 33.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

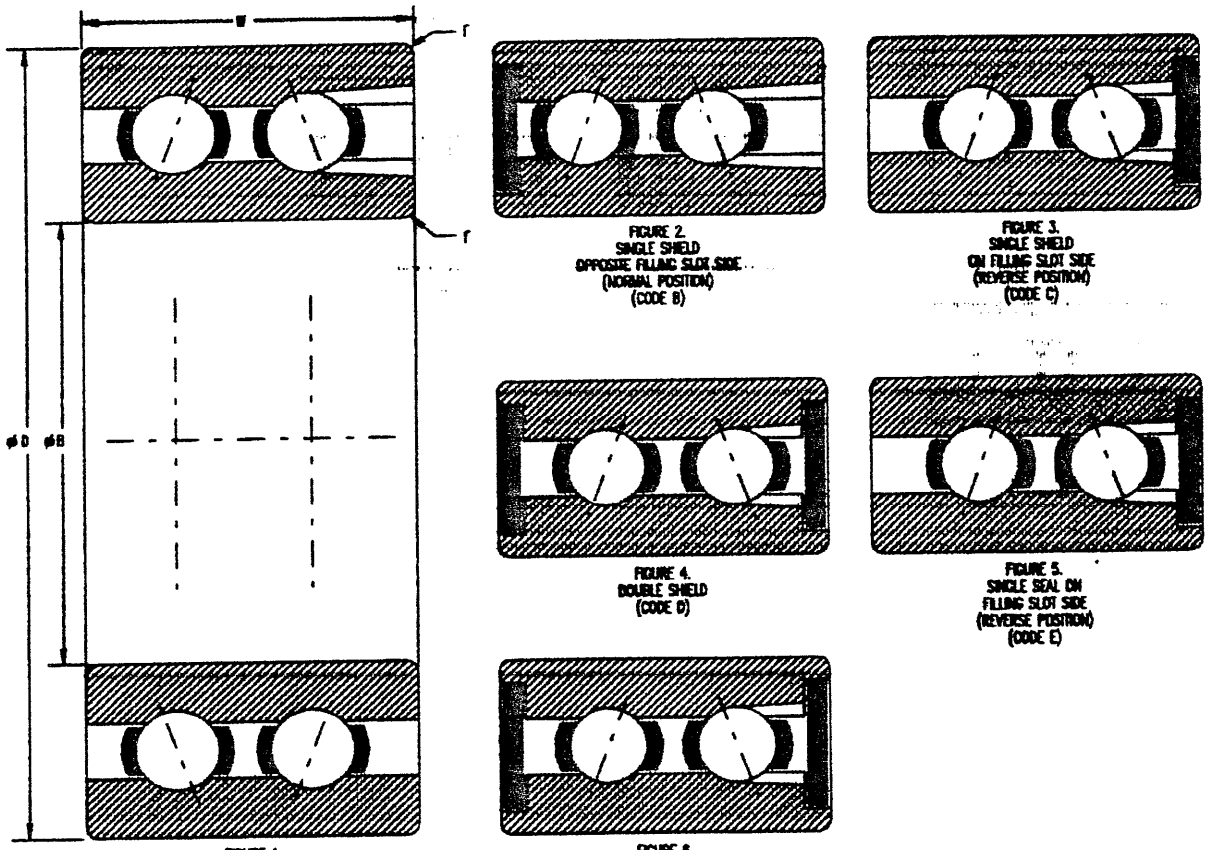


FIGURE 1. OPEN (CODE A)

FIGURE 2. SINGLE SHIELD OPPOSITE FILLING SLOT SIDE (NORMAL POSITION) (CODE B)

FIGURE 3. SINGLE SHIELD ON FILLING SLOT SIDE (REVERSE POSITION) (CODE C)

FIGURE 4. DOUBLE SHIELD (CODE D)

FIGURE 5. SINGLE SEAL ON FILLING SLOT SIDE (REVERSE POSITION) (CODE E)

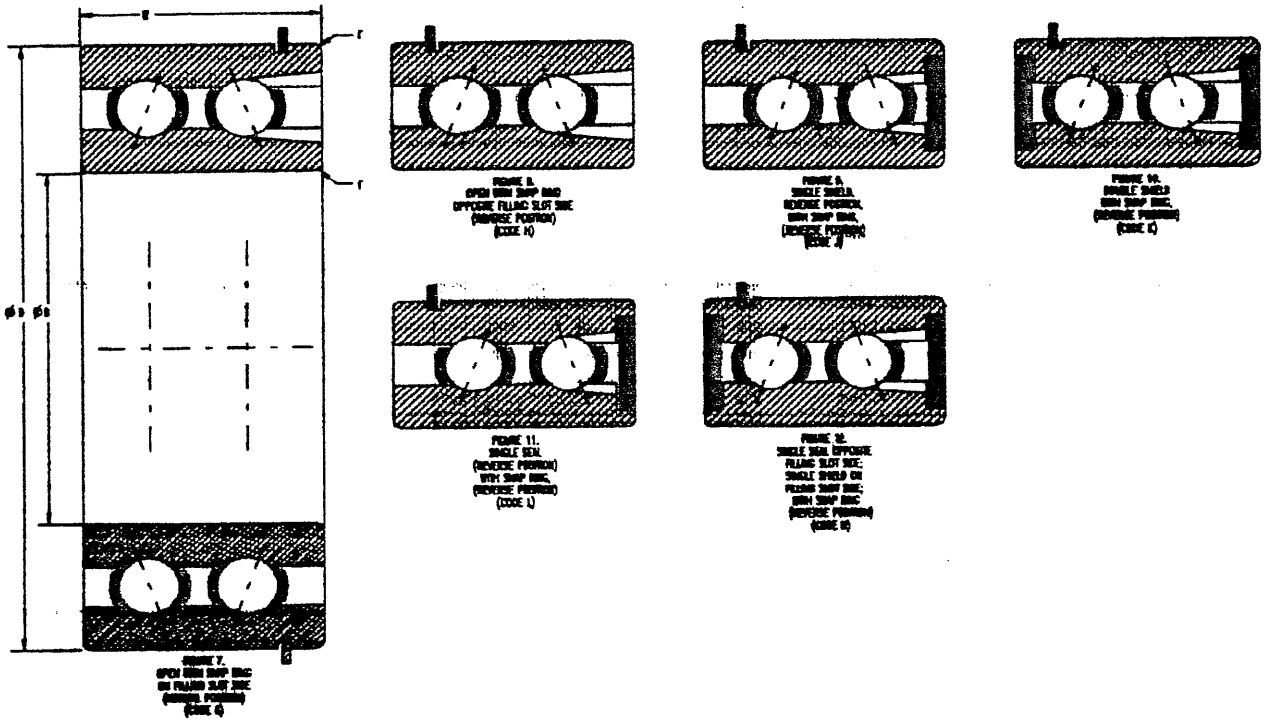
FIGURE 6. SINGLE SEAL OPPOSITE FILLING SLOT SIDE SINGLE SHIELD ON FILLING SLOT SIDE (CODE F)

BEARING CONFIGURATIONS

AMSC/NA FSC 3110

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BEARING CONFIGURATION (Continued)

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REQUIREMENTS:

1. **Boundary dimensions.** The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields, seals, **and snap** rings. Shields shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with AMSI/AFBMA Standard 20. The number and positioning of shields, seals, and snap rings required shall be specified in the part number by the codes in table III.

4. Radial Internal **Clearance.** The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

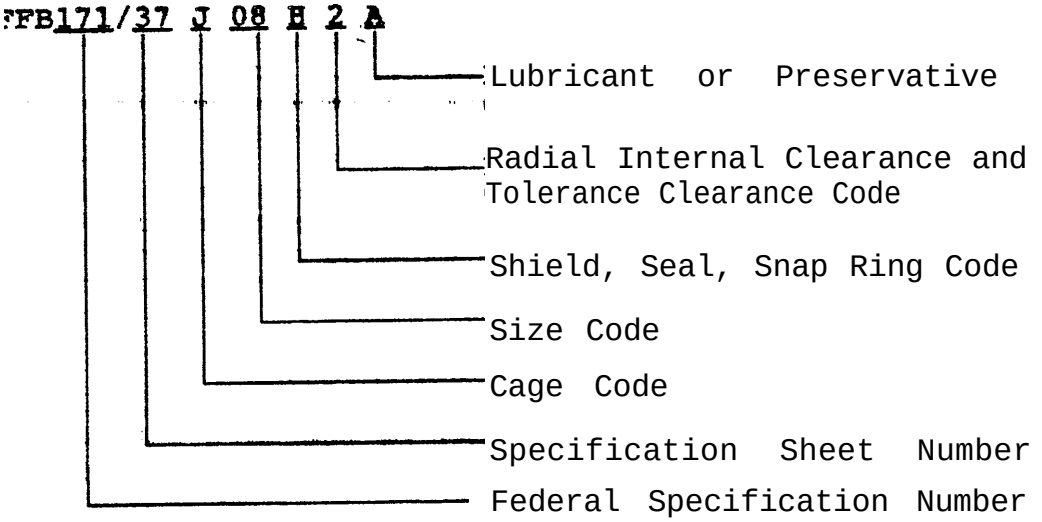
5. **Tolerance** Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. **Lubricant or Preservative Compound.** Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the “C” suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the “A" or “B” suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.

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7. **Part number.** The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields, seals, and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/37 J 08 H 2 A



FFB171/37J08H2A indicates - Bearing, cage-pressed steel or molded Plastic; bore 40mm; outside diameter 90mm; width 36.5mm; open (no shields or seals) with snap ring opposite filling slot side (reverse position); radial internal clearance symbol 0 (normal); tolerance class ABEC-1; grease in accordance with MIL-G-81322.

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TABLE I. Boundary dimensions.

Size Code 1/	OB Bore	OD Outside Diameter	W Width	Chamfer $r_s$min 2/	Shaft Shoulder Diameter Min 3/	C Dynamic load rating (pounds)
00	10	35	19.0	0.6	15	2110
01	12	37	19.0	1.0	18	2320
02	15	42	19.0	1.0	21	2660
03	17	47	22.2	1.0	23	3740
04	20	52	22.2	1.1	27	4740
05	25	62	25.4	1.1	32	6415
06	30	72	30.2	1.1	37	8310
07	35	80	34.9	1.5	44	10405
08	40	90	36.5	1.5	49	12725
09	45	100	39.7	1.5	54	14260
10	50	110	44.4	2.0	60	19035
11	55	120	49.2	2.0	65	22115
12	60	130	54.0	2.1	72	27080
13	65	140	58.7	2.1	77	28900
14	70	150	63.5	2.1	82	34465
15	75	160	68.3	2.1	87	37525
16	80	170	68.3	2.1	92	42890
17	85	180	73.0	3.0	99	46275
18	90	190	73.0	3.0	104	49725
19	95	200	77.8	3.0	109	53250
20	100	215	82.6	3.0	114	56870
21	105	225	87.3	3.0	119	60525
22	110	240	92.1	3.0	124	64265
24	120	260	106.0	3.0	134	71135
28	140	300	118.0	4.0	148	83275

1/ Size codes for Table I are industry bore code numbers.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximun shaft or housing fine? radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

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TABLE II. Codes for Cages.

Code	Cage
J	Pressed steel or molded plastic
M	Machined bronze or brass
T	Machined non-metallic (phenolic)
Y	Pressed brass

TABLE III. Codes for Shields and Seals.

Code	Shield and snap ring configuration
A	open (no shields) - fig.1
B	single shield opposite filling slot side, (normal position) - fig. 2
C	single shield on filling slot side, (reverse position) - fig. 3
D	double shield - fig. 4
E	open with snap ring on filling slot side, (normal position) - fig. 5
F	open with snap ring opposite filling slot side, (reverse position) - fig. 6
G	single shield, normal position, with snap ring, (normal position) - fig. 7
H	single shield, normal position, with snap ring, (reverse position) - fig. 8
J	single shield, reverse position, with snap ring, (normal position) - fig. 9
K	single shield, reverse position, with snap ring, (reverse position) - figure 10
L	double shield with snap ring, (normal position)-fig. 11
M	double shield with snap ring, (reverse position)-fig. 12

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

Code	Radial Internal Clearance	Tolerance Class
1	Symbol 2 (less than normal)	ABEC-1
2	Symbol 0 (normal)	
3	Symbol 3 (greater than normal)	
4	Symbol 4 (greater than symbol 3)	
5	Symbol 2 (less than normal)	ABEC-5
6	Symbol 0 (normal)	
7	Symbol 3 (greater than normal)	
8	Symbol 4 (greater than symbol 3)	

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TABLE V. Codes for Lubricant and Preservative Compound.

Code	Lubricant or Preservative Compound
A	Grease in accordance with MIL-G-81322
B	Grease in accordance with DOD-G-24508
C	Preservative compound in accordance with MIL-P-197

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

(Project 3110-0848-37)

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